UNDERGRADUATE BIOTECHNOLOGY / STRUCTURAL BIOINFORMATICS

Task 3: Protein Structure Prediction and Structural Feature Analysis of Human Insulin Precursor

Sequence annotations and a cautious interpretation of the AlphaFold Database model

Student _____ Course / section _____ Date _____

Abstract

This report examines the human insulin precursor using the reviewed UniProtKB entry P01308 and its corresponding AlphaFold Protein Structure Database record. The entry encodes a 110-amino-acid precursor containing a signal peptide, insulin B-chain region, connecting C-peptide, and insulin A-chain region. UniProt annotations also identify three disulfide bonds that connect the A and B chains and stabilize the mature hormone. The current AlphaFold record (model v6) reports a global confidence metric of 52.91; 87.3% of residues are classified as low or very low confidence. The structure is therefore useful as a predicted, sequence-linked model, but its detailed geometry—especially the precursor segments—must be interpreted cautiously. The report distinguishes the full precursor from proinsulin and mature insulin.

Introduction

Protein structure prediction estimates a three-dimensional (3D) arrangement from an amino-acid sequence. Predicted structures can help researchers formulate hypotheses about folding, molecular interactions, mutations, and protein engineering, particularly when experimental structures are unavailable. They do not replace experimental evidence: model confidence varies by region, and a predicted monomer may not represent processing, membrane context, oligomerization, or the mature biological product.

Protein Information

Attribute	Record
Protein name	Insulin precursor (INS_HUMAN; UniProt recommended name: Insulin)
Organism	*Homo sapiens* (human)
UniProtKB accession	P01308 (reviewed Swiss-Prot entry)
Sequence length	110 amino acids
Structure database	AlphaFold Protein Structure Database, AF-P01308-F1, model v6

Methodology

The UniProtKB P01308 record was consulted for the organism, sequence length, precursor features, and disulfide-bond annotations. The AlphaFold Database entry and its official API record were checked for model identity, sequence coverage, version, and confidence summary. The figure on page 2 is a Cα-trace rendering generated from the official AlphaFold model-v6 coordinate file and colored by UniProt precursor regions; it is not a screenshot of the web viewer. No PyMOL or Chimera visualization is claimed. Structural annotations and model confidence are reported separately.

Precursor organization and predicted model

The 110-residue database sequence corresponds to preproinsulin: the translated precursor before signal-peptide removal.

Precursor residues	Region	Biological interpretation
1–24	Signal peptide	Directs nascent preproinsulin into the secretory pathway; removed to form proinsulin.
25–54	B-chain region	Becomes the 30-residue B chain of mature insulin.
55–56	RR processing junction	Basic residues at the B-chain/C-peptide boundary.
57–87	Connecting C-peptide	Maintains connectivity during proinsulin folding; removed during maturation.
88–89	KR processing junction	Basic residues at the C-peptide/A-chain boundary.
90–110	A-chain region	Becomes the 21-residue A chain of mature insulin.

Disulfide bonds. UniProt annotates three cysteine pairings using full-precursor numbering: Cys31–Cys96 (B7–A7) and Cys43–Cys109 (B19–A20) are interchain links; Cys95–Cys100 (A6–A11) is intrachain within the A chain. These covalent bonds stabilize mature insulin. They are curated protein annotations, not measurements made from the AlphaFold rendering.

Structure Visualization

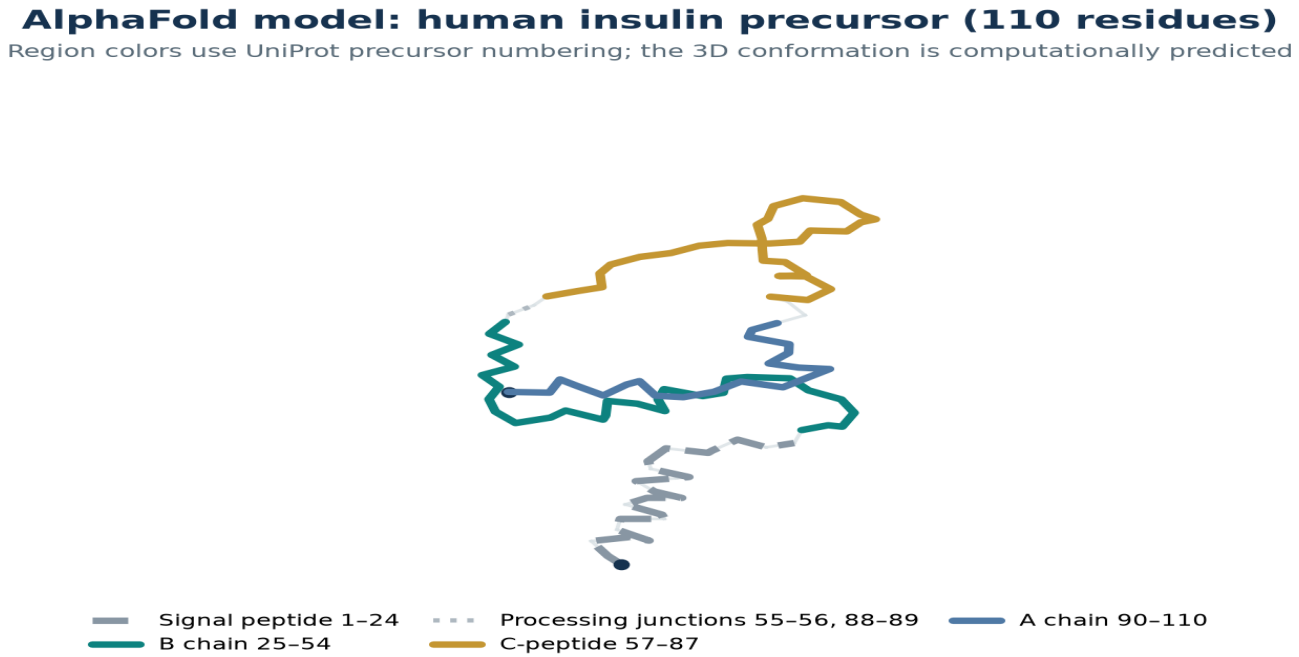


Figure 1. Predicted three-dimensional Cα trace for human insulin precursor (UniProt P01308), rendered from AlphaFold Database model AF-P01308-F1, v6. Colors denote UniProt-annotated precursor regions. The image is a coordinate rendering, not an experimentally determined structure and not a screenshot of the AlphaFold viewer. Given the low confidence summary, the displayed global arrangement should not be treated as a validated fold.



USER-SUPPLIED SCREENSHOT CROP — UNRELATED STRUCTURE
The source viewer identifies PDB 1TQN, human microsomal CYP3A4. This is not insulin (P01308) and must not be used as insulin evidence. Replace it with a screenshot from the AlphaFold P01308 viewer before submission.

INTERPRETATION / REFERENCES

Biological meaning and model limitations

Discussion

Insulin is synthesized as a single precursor polypeptide. The signal peptide supports entry into the secretory pathway and is removed first, producing proinsulin. Proinsulin then folds so that its A- and B-chain regions are linked by two interchain disulfide bonds; the C-peptide connects those regions during biosynthesis. Proteolytic processing removes C-peptide and the flanking basic residues, leaving mature insulin as the disulfide-linked A and B chains. Thus, the full 110-residue AlphaFold sequence is not the same molecule as mature insulin.

The AlphaFold Database record reports a global confidence metric of 52.91. Its residue-confidence distribution is 50.9% very low, 36.4% low, 12.7% confident, and 0% very high. These values were checked against the official P01308 API record (model v6). This is a low-confidence prediction overall, so the 3D trace is best treated as an illustrative model rather than evidence for precise distances, contacts, disulfide geometry, or the conformation of the processed hormone. AlphaFold predictions are computational models; they are not experimentally determined structures. Mature insulin's disulfide connectivity is supported by UniProt annotations and the cited literature, not inferred solely from this predicted precursor model.

Conclusion

UniProt P01308 describes a 110-amino-acid human insulin precursor with a signal peptide, B-chain region, C-peptide, A-chain region, and three annotated disulfide bonds. Signal-peptide removal and C-peptide processing produce mature insulin, whose A and B chains remain joined by two interchain disulfides and whose A chain contains one intrachain disulfide. The AlphaFold model provides a predicted structural view, but its low confidence requires caution and prevents strong claims about detailed 3D geometry.

References

1. UniProt Consortium. *INS_HUMAN / Insulin precursor, UniProtKB P01308*. https://www.uniprot.org/uniprotkb/P01308/entry (accessed 10 October 2026).
2. AlphaFold Protein Structure Database. *AF-P01308-F1: Insulin (Homo sapiens)*, model v6; entry and model-coordinate record. https://alphafold.ebi.ac.uk/entry/P01308 (accessed 10 October 2026). Coordinates: https://alphafold.ebi.ac.uk/files/AF-P01308-F1-model_v6.pdb
3. Varadi M, Anyango S, Deshpande M, et al. AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research*. 2022;50(D1):D439–D444. doi:10.1093/nar/gkab1061.
4. Jumper J, Evans R, Pritzel A, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*. 2021;596:583–589. doi:10.1038/s41586-021-03819-2.

Before submission

■ Add student name, course/section, and date on page 1.

■ Replace the CYP3A4 crop on page 2 with a student-captured screenshot of the official AlphaFold P01308 viewer, if required by the instructor.

■ Recheck the AlphaFold entry/version and confidence summary on the day of submission; database records can change.